Homework 20.11.26

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Exercise 40 (τ-nice sampling)

Let $S \subseteq \{1, \dots, n\}$ be chosen uniformly at random among all subsets of size τ, where $1 \leq \tau \leq n$. Prove that

$$p_i := \mathbb{P}(i \in S) = \frac{\tau}{n} \qquad \forall i = 1, \dots, n.$$

Proof. Since S is chosen uniformly, p_i is the number of size τ subsets that contain i divided by the number of all size τ subsets. The number of all size τ subsets $S \subseteq \{1, \dots, n\}$ is

$$\binom{n}{\tau} = \frac{n!}{\tau!\,(n-\tau)!}.$$

Now we compute how many of them contain i. Such a subset contains i and $\tau - 1$ other elements, which can be chosen arbitrarily from the remaining $n - 1$ elements, so there are

$$\binom{n-1}{\tau-1} = \frac{(n-1)!}{(\tau-1)!\,(n-\tau)!}$$

of them. Dividing this number by $\binom{n}{\tau}$, the factor $(n-\tau)!$ cancels, and using $n! = n\,(n-1)!$ and $\tau! = \tau\,(\tau-1)!$ we get

$$p_i = \frac{\binom{n-1}{\tau-1}}{\binom{n}{\tau}} = \frac{(n-1)!}{(\tau-1)!\,(n-\tau)!} \cdot \frac{\tau!\,(n-\tau)!}{n!} = \frac{(n-1)!}{n!} \cdot \frac{\tau!}{(\tau-1)!} = \frac{\tau}{n}. \qquad \square$$

Convergence of SGD-NICE

Setting and notation

Let $f = \frac{1}{n}\sum_{i=1}^n f_i$ with $n \geq 2$, and fix a minibatch size $\tau \in \{1, 2, \dots, n\}$. Let

$$n' := \binom{n}{\tau} \qquad \text{and} \qquad \mathcal{S} := \big\{\Delta \subseteq \{1, \dots, n\} \;:\; |\Delta| = \tau\big\} = \{\Delta_1, \dots, \Delta_{n'}\},$$

i.e. $\mathcal{S}$ is the set of all n' subsets of $\{1, \dots, n\}$ with τ elements. SGD-NICE is SGD with the gradient estimator

$$g(x) := \frac{1}{\tau}\sum_{i \in S} \nabla f_i(x),$$

where S is a τ-nice sampling, i.e. S is chosen uniformly from $\mathcal{S}$. We use the following constants:

$$A'' := \frac{n-\tau}{\tau(n-1)} \max_i L_i + \frac{n(\tau-1)}{\tau(n-1)} L,$$

$$\mathrm{Var}\big[g(x^\star)\big] = \frac{n-\tau}{\tau(n-1)} \left(\frac{1}{n}\sum_{i=1}^n \|\nabla f_i(x^\star)\|^2 - \|\nabla f(x^\star)\|^2 \right) \quad \text{(formula (79) of Lemma 45)},$$

$$A := 2A'', \qquad C := 2\mathrm{Var}\big[g(x^\star)\big].$$

Statement

Theorem (Convergence of SGD-NICE)**.** *Assume that f_i is convex and L_i-smooth for all i, and that f is μ-convex and L-smooth. Choose any relative error tolerance $0 < \delta < 1$ and the stepsize*

$$\gamma = \min\left\{\frac{1}{A},\ \frac{\mu\delta\left\|x^0 - x^\star\right\|^2}{C}\right\} = \min\left\{\frac{1}{2A''},\ \frac{\mu\delta\left\|x^0 - x^\star\right\|^2}{2\mathrm{Var}\left[g(x^\star)\right]}\right\}.$$

Then the iterates of SGD-NICE satisfy

$$k \geq \max\left\{\frac{2A''}{\mu},\ \frac{2\mathrm{Var}\left[g(x^\star)\right]}{\delta\mu^2\left\|x^0 - x^\star\right\|^2}\right\}\log\frac{1}{\delta} \quad \Rightarrow \quad \mathbb{E}\left[\left\|x^k - x^\star\right\|^2\right] \leq 2\delta\left\|x^0 - x^\star\right\|^2.$$

Equivalently, for an absolute accuracy $\varepsilon := 2\delta\left\|x^0 - x^\star\right\|^2$ (i.e. $\delta = \frac{\varepsilon}{2\|x^0 - x^\star\|^2}$),

$$k \geq \max\left\{\frac{2A''}{\mu},\ \frac{4\mathrm{Var}\left[g(x^\star)\right]}{\varepsilon\mu^2}\right\}\log\frac{2\left\|x^0 - x^\star\right\|^2}{\varepsilon} \quad \Rightarrow \quad \mathbb{E}\left[\left\|x^k - x^\star\right\|^2\right] \leq \varepsilon.$$

Proof

Proof. The idea is to use the methods from SGD without batches here in SGD with batches. The proof has three steps: we rewrite SGD-NICE as SGD-US, we apply Theorem 33, and we check the AC assumption.

1) SGD-NICE as SGD-US. For every $i \in \{1, \ldots, n'\}$ define

$$f_i^*(x) := \frac{1}{\tau}\sum_{j \in \Delta_i} f_j(x).$$

It is easy to notice that

$$f(x) = \frac{1}{n}\sum_{j=1}^{n} f_j(x) = \frac{1}{n'}\sum_{i=1}^{n'} f_i^*(x),$$

since, as in Exercise 40, every f_j appears in exactly $\binom{n-1}{\tau-1} = \frac{\tau}{n}n'$ of the functions f_i^*, so its total weight on the right is $\frac{1}{n'}\cdot\frac{1}{\tau}\cdot\frac{\tau}{n}n' = \frac{1}{n}$. Also it is easy to notice that the estimator of SGD-NICE can be also defined as in SGD-US:

$$g(x) = \nabla f_i^*(x), \qquad i \text{ chosen uniformly from } \{1, \ldots, n'\}.$$

So we have n' functions f_i^* with $f = \frac{1}{n'}\sum_{i=1}^{n'} f_i^*$ and $g(x) = \nabla f_i^*(x)$ for a uniformly chosen i, which is exactly the setup of SGD-US. In particular, g is unbiased: $\mathbb{E}\left[g(x)\right] = \frac{1}{n'}\sum_{i=1}^{n'}\nabla f_i^*(x) = \nabla f(x)$.

2) Applying Theorem 33. Since f is μ-convex and g is unbiased, if the AC assumption holds with constants A and C, then Theorem 33 says that for $0 < \gamma \leq \frac{1}{A}$

$$\mathbb{E}\left[\left\|x^k - x^\star\right\|^2\right] \leq (1 - \gamma\mu)^k\left\|x^0 - x^\star\right\|^2 + \frac{\gamma C}{\mu},$$

and from Commentary III of Theorem 33 we get

$$\mathbb{E}\left[\left\|x^k - x^\star\right\|^2\right] \leq 2\delta\left\|x^0 - x^\star\right\|^2$$

as long as

$$\gamma = \min\left\{\frac{1}{A},\ \frac{\mu\delta\left\|x^0 - x^\star\right\|^2}{C}\right\} \qquad \text{and} \qquad k \geq \frac{1}{\gamma\mu}\log\frac{1}{\delta}.$$

So if we show that g satisfies the AC assumption with $A = 2A''$ and $C = 2\mathrm{Var}\big[g(x^\star)\big]$, we get exactly the stepsize of the theorem and

$$\frac{1}{\gamma\mu} = \max\left\{\frac{A}{\mu},\ \frac{C}{\delta\mu^2\left\|x^0 - x^\star\right\|^2}\right\} = \max\left\{\frac{2A''}{\mu},\ \frac{2\mathrm{Var}\big[g(x^\star)\big]}{\delta\mu^2\left\|x^0 - x^\star\right\|^2}\right\},$$

which is what we have to prove. Substituting $\delta = \frac{\varepsilon}{2\|x^0 - x^\star\|^2}$ gives the second form with ε.

3) The AC assumption. It remains to prove that

$$G^k := \mathbb{E}\big[\left\|g^k - \nabla f(x^\star)\right\|^2 \ \big|\ x^k\big] \leq 2A\, D_f(x^k, x^\star) + C.$$

For this, from Lemma 45 we have

$$\mathbb{E}\big[\|g(x) - g(y)\|^2\big] \leq 2A''\, D_f(x, y).$$

As a result, applying Theorem 36 (with $C''(y) = 0$), we get that

$$G(x, y) \leq \underbrace{4A''}_{2A}\, D_f(x, y) + 2\mathrm{Var}\big[g(y)\big].$$

Taking $x = x^k$ and $y = x^\star$, this is exactly the AC assumption with $A = 2A''$ and $C = 2\mathrm{Var}\big[g(x^\star)\big]$, which finishes the proof. □